

Auteur: Mathijs Pittoors
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 27.17

$$\int \frac{x^4 + x^3 + 3x^2 + x + 1}{x^3 + 1} dx$$

Euclidische deling:

$$\begin{array}{r|l} \begin{array}{rrrrr} x^4 & +x^3 & +3x^2 & +x & +1 \\ -x^4 & & & -x & \end{array} & \begin{array}{rr} x^3 & +1 \\ x & +1 \end{array} \\ \hline \begin{array}{rrrr} x^3 & 3x^2 & & +1 \\ -x^3 & & & -1 \end{array} & \\ \hline \begin{array}{rr} 3x^2 & \end{array} & \end{array}$$

$$\Rightarrow \int x + 1 dx + \int \frac{3x^2 dx}{x^3 + 1} = \frac{x^2}{2} + x + \int \frac{3x^2 dx}{x^3 + 1}$$

Neem $t = x^3 + 1$ dan $dt = 3x^2 dx$ dus $dx = \frac{dt}{3x^2}$

$$\text{dan } \frac{x^2}{2} + x + \int \frac{dt}{t} = \frac{x^2}{2} + x + \ln |x^3 + 1| + c \text{ met } c \in \mathbb{R}$$

References